



Control your Mac using only hand gestures and your webcam — no hardware, no cost, no setup beyond Python. Built with MediaPipe, OpenCV, and PyAutoGUI.

Features

-  **Volume Control** — raise or lower system volume with a single finger
-  **Play / Pause** — control media playback with an open hand
-  **Tab Switching** — navigate browser or app tabs with gestures
-  **Brightness Control** — adjust screen brightness hands-free
-  **Screenshot** — take a screenshot with a gesture or press S
-  **Air Mouse** — move your cursor by pointing your index finger in the air
-  **Hold-to-Trigger** — gestures require a short hold to prevent accidental triggers
-  **Live Overlay UI** — real-time gesture guide and progress bar on the camera feed
-  **Optimised Performance** — frame skipping for smooth experience on M1/M2 Macs

Gesture Reference

Gesture	Action
 Index finger only	Volume Up
 Pinky finger only	Volume Down
 All fingers open	Play / Pause
 Peace sign (index + middle)	Next Tab
 Thumb only	Previous Tab
3 fingers (index + middle + ring)	Brightness Up
 Fist (all fingers closed)	Brightness Down
 Index + Pinky	Screenshot
 Thumb + Index	Air Mouse ON
 Fist (while in Air Mouse)	Air Mouse OFF

 **Tip:** Hold each gesture steady for ~0.5 seconds until the blue progress bar fills up — then it fires!

Tech Stack

Tool	Purpose
Python 3.12	Core language
MediaPipe	AI hand landmark detection
OpenCV	Webcam access & UI overlay
PyAutoGUI	Mouse, keyboard & media control
NumPy	Coordinate mapping for Air Mouse
osascript (built-in Mac)	Volume & brightness control
screencapture (built-in Mac)	Screenshot capture



Getting Started

Prerequisites

- Mac (M1 / M2 / M3 or Intel)
- Python 3.12 or higher
- A webcam (built-in or external)

Installation

Step 1 — Clone the repository

```
git clone https://github.com/YOUR_USERNAME/gesture-controller.git
cd gesture-controller
```

Step 2 — Install dependencies

```
pip3 install mediapipe==0.10.13 opencv-python pyautogui numpy
```

Step 3 — Download the MediaPipe hand model

```
curl -O https://storage.googleapis.com/mediapipe-models/hand_landmarker/hand_landmarker/float16/1/hand_landmarker.task
```

Step 4 — Run the app

```
python3 main.py
```

Mac Permissions Required

On first run, macOS will ask for the following permissions. Make sure to allow all of them:

Permission	Why It's Needed
Camera	To see your hand gestures
Screen Recording	To take screenshots
Accessibility	To control volume, brightness, and tabs

To grant permissions manually:

1. Go to **System Settings** → **Privacy & Security**
2. Enable **Camera**, **Screen & System Audio Recording**, and **Accessibility** for Terminal

Project Structure

```
gesture-controller/
├── main.py           ← Main application
└── hand_landmarker.task ← MediaPipe AI model file
```

Air Mouse Mode

Air Mouse lets you control your cursor with your index finger:

1. Make the **Thumb + Index** gesture to activate
2. Point your **index finger** at the screen — the cursor follows!
3. Make a **Fist** to exit Air Mouse mode

The cursor uses smoothing so it does not shake — the more still your hand, the more precise the cursor!

Keyboard Shortcuts

Key	Action
S	Take a screenshot immediately

Key Action

Q Quit the app

Customising Gestures

You can remap any gesture by editing the `GESTURE_MAP` dictionary at the top of `main.py`:

```
GESTURE_MAP = {  
    (0, 1, 0, 0, 0): "VOLUME UP",    # index finger up  
    (0, 0, 0, 0, 1): "VOLUME DOWN",  # pinky finger up  
    (1, 1, 1, 1, 1): "PLAY PAUSE",   # all fingers up  
    # Add or change mappings here!  
}
```

Each tuple represents fingers in order: (thumb, index, middle, ring, pinky) where 1 = up and 0 = down.

Troubleshooting

Problem	Fix
No module named cv2	Run <code>pip3 install opencv-python</code>
module mediapipe has no attribute solutions	Run <code>pip3 install mediapipe==0.10.13</code>
Camera not opening	Allow Camera access in System Settings → Privacy & Security
Screenshot not saving	Allow Screen Recording in System Settings → Privacy & Security
Gestures triggering too easily	Increase <code>HOLD_FRAMES</code> value in <code>main.py</code> (default: 12)
App running slowly	Increase <code>PROCESS_EVERY</code> value in <code>main.py</code> (default: 2)

Credits

- Hand tracking powered by [MediaPipe](#) by Google

- Webcam processing via [OpenCV](#)

License

This project is open source and available under the [MIT License](#).

Built with love as a software engineering project. No hardware required — just your hand and a webcam.